# Module 9 Assignment — Computational Lab 9

Start Assignment

- Due Thursday by 6pm
- Points 100
- Submitting a text entry box, a website url, or a file upload
- Available Sep 11 at 12am - Dec 20 at 11:59pm

**BIOT 6900 · AI-POWERED BIOTECH WORKFLOWS**

**Module 9 Assignment — Clinical NLP on Your Own Disease**

Week 9 · Part III · Computational Lab 9 · Fall 2026

**Weight:** 100 points · Computational Lab 9 (3% of course grade; your lowest lab is dropped).

## What this assignment is

You will run the clinical-NLP pipeline from this module on a disease of your own choosing, and then do the part that matters: look at what your model learned and separate genuine early signal from leakage. You design the clinical picture, the notebook builds a synthetic cohort and notes from it, extracts features, trains an outcome model, and reports feature importance. **It also injects two clinical features you did not configure, and you must diagnose each.** Everything you need is in the starter notebook, BIOT6900_Module9_Notebook3_Graded.ipynb, on Canvas.

**Maps to:** LO1 (apply AI tools to biological data), LO3 (interpret results in clinical context), and LO4 (evaluate model performance with proper metrics).

### The one idea you are graded on

**A feature can predict the outcome perfectly and still be worthless,** because it is caused by the outcome rather than preceding it. Telling signal from leakage, and defending the call, is the 40%.

## Before you start

- **Do the warm-ups first.** Run Notebook 1 (toy) and Notebook 2 (real workflow) before this one. They teach the exact moves you will be graded on here.

- The notebook uses only pandas and scikit-learn, already in your biot6900 environment. No clinical-NLP installs are needed.
- All patients and notes are synthetic and generated inside the notebook from your configuration. Nothing to download.

# What to do

Open the notebook and complete the cells marked TODO.

# Part A — Configure your disease

Replace the example CONFIG with your own disease. Put **2 to 3 genuine early signals** (findings that are **causes**, present before a patient deteriorates) in signals, and **2 to 3 consequences** (interventions or decisions taken **because** a patient is deteriorating) in consequences. Give each a few synonyms. Set SEED to your student-ID number.

**Hint.** A cause belongs in signals; a treatment, escalation, or end-of-life decision belongs in consequences. Getting this split right is clinical reasoning, and it is graded.

# Part B — Extract features

Run the extraction. It builds a feature for each of your concepts plus the two injected wildcards, across all 400 patients. Check the counts are sensible, and note the naive-versus-assertion number for one of your signals.

# Part C — Model and importance

Set declared_leak to the consequences you configured (not the wildcards yet). Run the cross-validated AUROC and the permutation-importance plot. Expect the all-features score to look high.

# Part D — Catch the leakage and interpret  (the 40%)

Answer D1 to D4 in the markdown cells, using the importance plot and the AUROC numbers as evidence and clinical reasoning as the argument.

- List the top features and classify each as signal or leakage, with a one-line clinical reason.
- Diagnose the two injected features (lactate_not_clearing, renal_replacement) as signal or leakage and justify. Add any leak you find to final_leak and re-run.
- Report the AUROC with everything, with declared leakage removed, and with all leakage removed. Interpret the drops.

- Would your genuine signals be documented early enough to act on? Name one way this synthetic cohort is unlike real data.

**Hint.** If your 'declared leakage removed' AUROC is still high, you missed a leak. That is the clue for D2.

# What to submit

- **Your completed notebook,** run top to bottom (Kernel → Restart & Run All), committed to your biot6900 repo.
- **Your written D1 to D4 answers** in the markdown cells.
- **A short README** naming your disease and anything that did not work.

**Then post your repo link on Canvas.** Common mistake: committing but forgetting to push. Check your files appear on github.com before you post.

# Due date & late policy

**Due Thursday, November 12 at 6:00 pm** (start of Week 10). Per the syllabus: –10% up to 1 day late, –20% up to 2 days, –50% up to 3 days, and no credit after the 3rd day unless arranged in advance.

# How it's graded

Course 60 / 40 split: **60 points execution** (working pipeline, correct method) and **40 points interpretation and communication** (understanding what the model did).

**Partial credit** is awarded for a correct approach even with minor technical errors. If a step would not run, say what you were trying to do and what happened. Explaining your reasoning earns credit.

# Getting help

- **Office hours** with your laptop are the fastest way to unstick anything.
- **The Notebook 2 lab guide** has a troubleshooting table that applies here too.
- **Email** with your name, the class title, and the exact error text or a screenshot. Allow up to 48 hours.

*BIOT 6900 · Module 9 Assignment · Fall 2026*

**Computational Lab 9**

| Criteria | Ratings | | Pts |
|---|---|---|---|
| Disease configured sensibly; notebook runs clean (execution) | **20 pts**<br>**Full Marks**<br>2–3 signals and 2–3 consequences with synonyms; causes vs interventions sorted correctly; Restart & Run All works. | **0 pts**<br>**No Marks** | 20 pts |
| Extraction correct (execution) | **20 pts**<br>**Full Marks**<br>Features built across the cohort; naive-vs-assertion check reported and sensible. | **0 pts**<br>**No Marks** | 20 pts |
| Model and importance produced (execution) | **20 pts**<br>**Full Marks**<br>Cross-validated AUROC, permutation importance, and the leakage-removal comparison all run. | **0 pts**<br>**No Marks** | 20 pts |
| D1–D2: signal vs leakage, incl. the two wildcards (interpretation) | **22 pts**<br>**Full Marks**<br>Top features classified with clinical reasoning; both injected features diagnosed and defended; the hidden leak caught. | **0 pts**<br>**No Marks** | 22 pts |
| D3: drops quantified and interpreted (interpretation) | **8 pts**<br>**Full Marks**<br>The three AUROC numbers reported, with a correct reading of what each drop means. | **0 pts**<br>**No Marks** | 8 pts |
| D4 + documentation (communication) | **10 pts**<br>**Full Marks**<br>Reality check answered; README present; a grader could re-run it. | **0 pts**<br>**No Marks** | 10 pts |

Total Points: 100